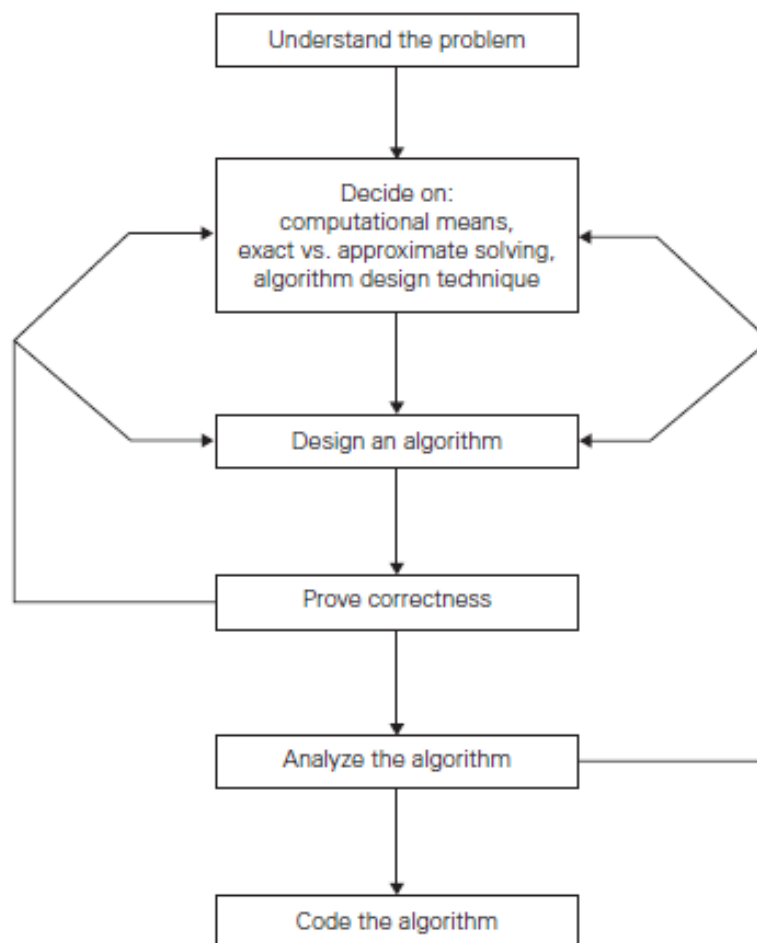


Text Book:
Introduction to the Design and Analysis of Algorithms
Author: Anany Levitin
2nd Edition

Fundamentals of Algorithmic Problem Solving



Understand the Problem

- Algorithms are procedural solutions to problems.
- An input to an algorithm specifies an instance of the problem the algorithm solves.
- Boundary conditions should be clearly understood

Decide on computational means

- Sequential vs Parallel algorithm
- Exact vs Approximation algorithm
- Data Structures + Algorithms = Programs

Design Algorithm

Specifying algorithm

- Natural Language
- Pseudo code
- Flowchart

Correctness:

Mathematical Induction

Exact Algorithms: correct algorithm is the one that works for all legitimate inputs.

Approximate Algorithms: Error in tolerance limit

Analyzing Algorithm

- Time vs Space efficiency
- Simplicity vs Generality

Coding the algorithm

- Testing
- Debugging
- Code optimization